

Mathematical Modeling of Ion Transport and Water Dissociation at the Ion-Exchange Membrane/Solution Interface in Intense Current Regimes

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Abstract—At current densities exceeding the limiting current density, H^+ and OH^- ions are generated at the interface of the ion-exchange membrane with a depleted solution as a result of the dissociation of water molecules. At present, it is generally accepted that water splitting occurs in a thin (a few nanometers) layer inside the membrane, this reaction being catalytic in nature. The mathematical model of ion transport in the diffusion layer near the membrane surface has been constructed and numerically studied under conditions when dissociation and recombination processes involving water molecules and H^+ and OH^- ions occur simultaneously. It has been shown that in overlimiting current regimes under very high voltages, intense noncatalytic dissociation of water molecules in the extended space charge region of the depleted solution can occur irrespective of the catalytic splitting of water. Since this region has macroscopic dimensions, the rate of noncatalytic water dissociation is comparable with the rate of the corresponding catalytic process. The obtained results significantly supplement modern concepts of the mechanism of generation of H^+ and OH^- ions in membrane systems, showing that this process can proceed not only in accordance with the conventional mechanism with the catalytic participation of functional groups and/or other compounds, but also via the noncatalytic mechanism that has not been taken into account to the present.

Keywords: ion-exchange membrane, limiting current, modeling, water dissociation, catalytic reaction, non-catalytic reaction

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INTRODUCTION

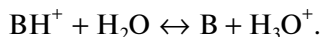
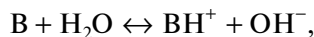
In intense current regimes when the current used is several times higher than the limiting electrodiffusion current, secondary (or coupled) concentration polarization phenomena arise in membrane systems [1]: the space electric charge occupies a macroscopic region comparable to the diffusion layer thickness, the pH of the solution undergoes a shift [2, 3] explained by the splitting of water, microconvective flows are formed, etc.

Taking into account the effect of the dissociation/recombination reaction of water molecules is important for understanding the transport processes in electromembrane systems, since some researchers believe that the appearance of the new charge carriers H^+ and OH^- can lead to a decrease or even disappearance of the space charge, which forms the basis of other transport mechanisms, for example, electroconvection.

The splitting of water in systems with ion-exchange membranes was first observed by Frilette [2] and Kressman and Tye [3] and has since been studied by a number of investigators [4–21].

Beginning with the works of Greben et al. [10] and Simons [11–13], it is generally assumed that water splitting occurs in thin surface layers of membranes, with this reaction being catalytic in nature [23]. Functional groups of the membrane or other compounds (precipitates of sparingly soluble salts or catalysts immobilized for a special purpose on the interface) are involved in this reaction. The assumption is substantiated by the following argument [12, 13]. If the H^+ and OH^- ions were generated within a thin reaction layer of a thickness δ_r at the usual, constant water dissociation rate characteristic of a free solution, then their fluxes would not exceed $k_d C_{H_2O} \delta_r$, where k_d is the rate constant of water dissociation and C_{H_2O} is the concentration of water in this layer. Assuming that $k_d = 2 \times 10^{-5} \text{ s}^{-1}$ at 25°C , $C_{H_2O} = 10 \text{ mol/L}$ (inside the membrane phase), and $\delta_r = 10 \text{ nm}$, the upper limit for the fluxes of water ions can be estimated at $2 \times 10^{-9} \text{ mol m}^{-2} \text{ s}^{-1}$. At the same time, their actual fluxes reach $5 \times 10^{-4} \text{ mol m}^{-2} \text{ s}^{-1}$ at 25°C for ion exchange membranes [12, 22]. In order to overcome this divergence, Simons [11–13] suggested that the water dissociation reaction is catalytic

in character. In the case when functional groups (B) of an anion-exchange membrane take part in this reaction, the generation process can be represented as a reaction proceeding in two steps [13, 14]:



The first reaction proceeds rapidly; the rate-determining step is the dissociation of the protonated functional group BH^+ . However, the rate constant of the second reaction is many orders of magnitude larger than the water dissociation rate constant k_d , a fact that explains the acceleration of the entire process of generation of H^+ and OH^- ions as a whole.

Later, this theory based on the assumption that water splitting is a catalytic process and occurs only in a thin boundary layer of the membrane was developed in many studies [14–17, 23], becoming generally accepted to date.

The theoretical analysis carried out in this paper shows that in an overlimiting current regime at a high potential drop, another scenario of the generation of H^+ and OH^- ions is possible, which is based on the intensive dissociation of water molecules in an extended space charge region. Since the size of this macroscopic region can be four orders of magnitude larger than the thickness of the reaction layer, the flux of H^+ and OH^- ions may also be as many times as large. Thus, these results substantially supplement modern ideas about the mechanism of generation of H^+ and OH^- ions in membrane systems, showing that this process can occur not only via the conventional catalytic mechanism, but also by a noncatalytic mechanism that is not currently taken into account.

In this study based on the results obtained in [18], we developed a mathematical model for the electrodiffusion of four types of ions simultaneously (two salt ions and H^+ and OH^- ions) in the diffusion layer near an ideally selective membrane with the combined effect of electroneutrality violation and the dissociation/recombination reaction of water molecules. A numerical analysis of the model in accordance with [24] made it possible to reveal the main features of the influence of the dissociation/recombination reaction on the transport of salt ions.

PHYSICAL STATEMENT OF THE PROBLEM

The one-dimensional steady-state transport of singly charged salt ions in the Nernst diffusion layer with allowance for water dissociation/recombination is described by the system of equations (1)–(7).

$$j_1^{(u)} = -\frac{F}{RT} D_1^{(u)} C_1^{(r)} \frac{d\phi^{(u)}}{dx^{(u)}} - D_1^{(u)} \frac{dC_1^{(u)}}{dx^{(u)}}, \quad (1)$$

$$x^{(u)} \in (0, \delta),$$

$$j_2^{(u)} = \frac{F}{RT} D_2^{(u)} C_2^{(u)} \frac{d\phi^{(u)}}{dx^{(u)}} - D_2^{(u)} \frac{dC_2^{(u)}}{dx^{(u)}}, \quad (2)$$

$$j_3^{(u)} = -\frac{F}{RT} D_3^{(u)} C_3^{(r)} \frac{d\phi^{(u)}}{dx^{(u)}} - D_3^{(u)} \frac{dC_3^{(u)}}{dx^{(u)}}, \quad (3)$$

$$j_4^{(u)} = \frac{F}{RT} D_4^{(u)} C_4^{(u)} \frac{d\phi^{(u)}}{dx^{(u)}} - D_4^{(u)} \frac{dC_4^{(u)}}{dx^{(u)}}, \quad (4)$$

$$\frac{d}{dx^{(u)}} \left(\frac{d\phi^{(u)}}{dx^{(u)}} \right) = -\frac{F}{\epsilon_0^{(u)}} (C_1^{(u)} - C_2^{(u)} + C_3^{(u)} - C_4^{(u)}), \quad (5)$$

$$0 = -\frac{dj_i^{(u)}}{dx^{(u)}} + R_i^{(u)}, \quad i = 1, \dots, 4, \quad (6)$$

$$R_1^{(u)} = R_2^{(u)} = 0, \quad R_3^{(u)} = R_4^{(u)} = k_d^{(u)} C_{H_2O}^{(u)} - k_r^{(u)} C_3^{(u)} C_4^{(u)} = k_r^{(u)} (k_W^{(u)} - C_3^{(u)} C_4^{(u)}). \quad (7)$$

Here, $k_d^{(u)} = 2 \times 10^{-5} \text{ s}^{-1}$ is the water dissociation rate constant; $C_{H_2O}^{(u)} = 55.6 \text{ mol/L}$ is the water concentration in the solution; $k_r^{(u)} = 1.1 \times 10^{11} \text{ L mol}^{-1} \text{ s}^{-1}$ is the recombination rate constant of hydrogen H^+ and hydroxyl OH^- ions; $k_W^{(u)} = \frac{k_d^{(u)} C_{H_2O}^{(u)}}{k_r^{(u)}} = 10^{-14} \text{ mol}^{-2} \text{ L}^{-2}$ is

the equilibrium constant at 25°C ; $\epsilon_0^{(u)}$ is the dielectric permittivity of the solution; F is Faraday's number, subscripts $i = 1$ and 2 refer to salt ions, $i = 3$ and $i = 4$ refer to hydrogen H^+ and hydroxyl OH^- ions, respectively; $E^{(u)} = -\frac{d\phi^{(u)}}{dx^{(u)}}$ is the electric field strength; $C_i^{(u)}$, $j_i^{(u)}$, $D_i^{(u)}$ are respectively the concentration, flux, and diffusion coefficient of the i th ion; and $I^{(u)}$ is the total current density, wherein

$$I^{(u)} = I_S^{(u)} + I_W^{(u)}, \quad \text{where} \quad (8)$$

$$I_S^{(u)} = F(j_1^{(u)} - j_2^{(u)}), \quad I_W^{(u)} = F(j_3^{(u)} - j_4^{(u)}). \quad (9)$$

$I_S^{(u)}$ is the density of the partial current due to salt ions, and $I_W^{(u)}$ is the density of the partial current due to water ions. Superscript (u) means that the corresponding quantities are dimensional:

From Eqs. (6), (7), and (9) it follows that:

$$j_1^{(u)} = \text{const}, \quad j_2^{(u)} = \text{const},$$

$$I_S^{(u)} = F(j_1^{(u)} - j_2^{(u)}) = \text{const}, \quad I_W^{(u)} = \text{const}, \quad (10)$$

but $j_3^{(u)} \neq \text{const}$, $j_4^{(u)} \neq \text{const}$.

We assume that $x^{(u)} = 0$ corresponds to the depth of the solution, where the electroneutrality condition $C_1^{(u)} - C_2^{(u)} + C_3^{(u)} - C_4^{(u)} = 0$, is satisfied and $x^{(u)} = \delta^{(u)}$ corresponds to the conditional solution/membrane

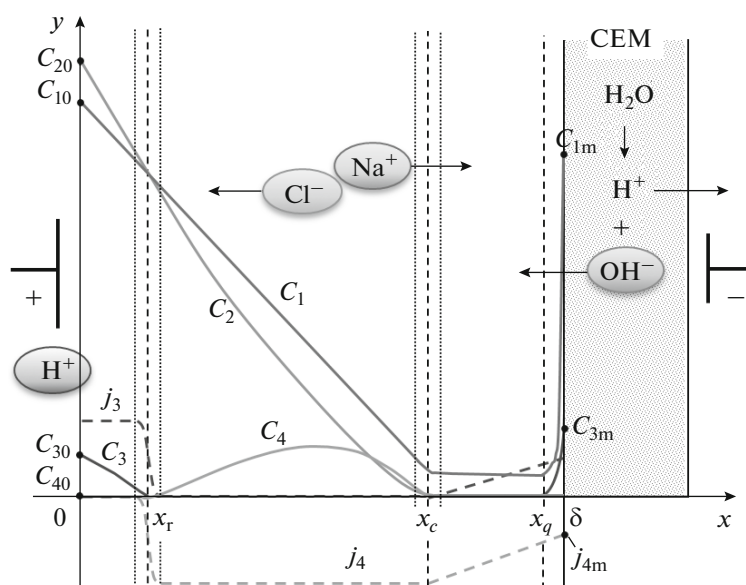


Fig. 1. Scheme of the structure of the diffusion layer under intense current regimes. The data points at $x = 0$ and $x = \delta$ are the preset boundary values of the functions sought. The scheme is valid both in dimensional and dimensionless forms. In the latter case, $\delta = 1$. (Not to the scale.)

interface. Thus, $\delta^{(u)}$ is the thickness of the Nernst diffusion layer.

In this paper, we consider for definiteness a cation-exchange membrane (CEM) (Fig. 1), which is assumed to be ideally selective ($j_2^{(u)} = 0$).

We will take that in the bulk of the solution ($x^{(u)} \leq 0$), the concentrations of all the ions are given so that $C_3^{(u)}(0) \gg C_4^{(u)}(0)$, $C_4^{(u)}(0) \approx 0$. Note that $\text{pH} < 7$. Thus, it may be assumed that $C_3^{(u)}(0) = C_{30}^{(u)} > 0$, $C_4^{(u)}(0) = 0$, $C_1^{(u)}(0) = C_{10}^{(u)} > 0$, and $C_2^{(u)}(0) = C_{20}^{(u)} > 0$.

In the bulk of the solution, the local electroneutrality condition is fulfilled with great accuracy. For boundary values, this means that $C_{10}^{(u)} + C_{30}^{(u)} = C_{20}^{(u)}$. In an overlimiting current regime, a space charge region (SCR) appears near $x^{(u)} = \delta^{(u)}$, where the local electroneutrality condition is not satisfied and Poisson equation (5) should be used to describe the transport process [25]. Its use, in turn, requires defining boundary conditions at $x^{(u)} = \delta^{(u)}$. As such conditions, we set values of the cation concentrations $C_1^{(u)}(\delta) = C_{1m}^{(u)}$, $C_3^{(u)}(\delta) = C_{3m}^{(u)}$, the sum of which is on the order of the exchange capacity of the cation-exchange membrane. As noted above, in the thin boundary layer of the cation-exchange membrane, the splitting of water via the catalytic mechanism is possible. At the membrane/solution interface, an OH^- ion flux is formed, directed into the bulk of the depleted solution, so the catalytic dissociation of water can be taken into

account by setting the OH^- ion flux density at the point $x^{(u)} = \delta^{(u)}$: $j_4^{(u)}(\delta) = j_{4m}^{(u)}$.

It is shown in this paper that the diffusion layer $[0, \delta^{(u)}]$ (Fig. 1) consists of the electroneutrality region (ENR) (interval $[0, x_c^{(u)})$) and the space charge region (SCR) (interval $(x_c^{(u)}, \delta^{(u)})$). In the ENR, there is a narrow region of recombination (RR) near the point $x_r^{(u)}$. Near the point $x_c^{(u)}$, there is a narrow intermediate region (IR) between the ENR and the SCR. The SCR consists of a nonequilibrium (extended) part $(x_c^{(u)}, x_q^{(u)})$ and a quasi-equilibrium part (diffuse electrical double layer, EDL).

Note that although a sodium chloride solution is considered below, the study is valid for a solution of an arbitrary binary salt.

TRANSITION TO A DIMENSIONLESS FORM

Let us take as a characteristic concentration $C_0^{(u)}$ the concentration of salt cations $C_{10}^{(u)}$. We denote the diffusion coefficient of the electrolyte (NaCl) as $D_0^{(u)}$ and proceed to the following dimensionless quantities:

$$x = \frac{x^{(u)}}{\delta^{(u)}}, \quad E = -\frac{\delta^{(u)} F}{RT} \frac{d\phi^{(u)}}{dx^{(u)}}, \quad I = \frac{I^{(u)} \delta^{(u)}}{D_0^{(u)} C_{10}^{(u)} F},$$

$$I_S = \frac{I_S^{(u)} \delta^{(u)}}{D_0^{(u)} C_{10}^{(u)} F}, \quad I_W = \frac{I_W^{(u)} \delta^{(u)}}{D_0^{(u)} C_{10}^{(u)} F}, \quad C_i = \frac{C_i^{(u)}}{C_{10}^{(u)}},$$

$$j_i = \frac{j_i^{(u)} \delta^{(u)}}{D_i^{(u)} C_{10}^{(u)}}, D_i = \frac{D_i^{(u)}}{D_0^{(u)}}, i = 1, 2, 3, 4, k_W = \frac{k_W^{(u)}}{(C_{10}^{(u)})^2} =$$

$$\frac{C_3^{(u)}(0) C_4^{(u)}(0)}{(C_{10}^{(u)})^2} = \left(\frac{C_3^{(u)}(0)}{C_{10}^{(u)}} \right)^2 = \left(\frac{C_4^{(u)}(0)}{C_{10}^{(u)}} \right)^2 \ll 1,$$

$$\varepsilon = \frac{RT \varepsilon_0^{(u)}}{C_{10}^{(u)} F^2 \delta^{(u)2}} = 2 \left(\frac{l_d^{(u)}}{\delta^{(u)}} \right)^2 \ll 1, \text{ where } l_d^{(u)} = \sqrt{\frac{RT \varepsilon_0^{(u)}}{2 C_{10}^{(u)} F^2}},$$

$$a = \frac{RT \varepsilon_0^{(u)}}{F^2} \times \frac{k_r^{(u)}}{D_0^{(u)}} = \frac{RT \varepsilon_0^{(u)}}{2 C_{10}^{(u)} F^2} \times \frac{2 k_r^{(u)} C_{10}^{(u)}}{D_0^{(u)}} = \left(\frac{l_d^{(u)}}{l_r^{(u)}} \right)^2,$$

$$\text{where } l_r^{(u)} = \sqrt{\frac{D_0^{(u)}}{2 k_r^{(u)} C_{10}^{(u)}}}.$$

The total current density will be written as $I_S + I_W = I$, where

$$I_S = D_1 j_1 - D_2 j_2, \quad (11)$$

$$I_W = D_3 j_3 - D_4 j_4, \quad (12)$$

with j_2 being 0 for an ideally selective membrane; therefore, $I_S = D_1 j_1$.

Parameter k_W is the square of the ratio of the concentration of H^+ or OH^- ions in the neutral solution (pH 7) to the electrolyte concentration. Parameter ε is the doubled square of the ratio of the Debye length to the diffusion layer width $\delta^{(u)}$. Parameter a is the square of the ratio of the Debye length to the width of the recombination region $l_r^{(u)}$.

At real values of the concentration of the solution $C_{10}^{(u)}$ (in the range of $10^{-5} - 1$ mol/L) and the diffusion layer thickness $\delta^{(u)}$, the parameters ε and k_W can be considered as small quantities, since their values range from 10^{-11} to 10^{-6} and from 10^{-14} to 10^{-4} , respectively. The parameter a is on the order of 10; so, for example,

$$a \approx 12.92 \quad \text{for} \quad k_r^{(u)} = 1.112 \times 10^8 \frac{\text{m}^3}{\text{mol s}},$$

$$D_0^{(u)} = 1.61 \times 10^{-9} \frac{\text{m}^2}{\text{s}}, \quad \varepsilon_0^{(u)} = 7.172 \times 10^{-10} \frac{\text{A}^2 \text{s}^4}{\text{m}^3 \text{kg}},$$

$$T^{(u)} = 293.15 \text{ K}.$$

The dimensionless boundary value problem describing the model has the form:

$$\frac{dC_1}{dx} = C_1 E - j_1, \quad (13)$$

$$\frac{dC_2}{dx} = -C_2 E - j_2, \quad (14)$$

$$\frac{dC_3}{dx} = C_3 E - j_3, \quad (15)$$

$$\varepsilon D_3 \frac{d}{dx} j_3 = a(k_W - C_3 C_4), \quad (16)$$

$$\frac{dC_4}{dx} = -C_4 E - j_4, \quad (17)$$

$$\varepsilon D_4 \frac{d}{dx} j_4 = a(k_W - C_3 C_4), \quad (18)$$

$$\varepsilon \frac{dE}{dx} = C_1 - C_2 + C_3 - C_4. \quad (19)$$

$$\text{At } x = 0: C_1|_{x=0} = 1, C_2|_{x=0} = 1 + C_{30}, \quad (20)$$

$$C_3|_{x=0} = C_{30}, C_4|_{x=0} = C_{40} = 0.$$

$$\text{At } x = 1: C_1|_{x=1} = C_{1m}, C_3|_{x=1} = C_{3m}, \quad (21)$$

$$j_4|_{x=1} = j_{4m}.$$

Note. By continuity, the relationship $I_W = D_3 j_3 - D_4 j_4 = \text{const}$ must be valid up to the boundary. Hence, the following relation is true

$$I_W = D_3 j_{3m} - D_4 j_{4m} = D_3 j_{30} - D_4 j_{40}. \quad (22)$$

The thickness of the diffusion layer does not depend on electric current density or voltage. The increase in the current of salt ions is due to a decrease in the thickness of the electrically neutral part of the diffusion layer. The problem is a generalization of the Rubinstein–Shtilman problem [19] and the Kharkats problem [21]. The current density of salt cations can be calculated using the following formula, which is an approximate solution of the problem posed [20]:

$$I_1 = \frac{2 D_1 C_{10}}{\delta_{\text{ENR}}} + \frac{D_1}{D_3} I_3,$$

where $\delta_{\text{ENR}} \approx x_c$ is the width of the electroneutrality region. The first term is due to diffusion and migration of counterions in ENR, and the second term is due to the exaltation current [21].

MAIN RESULTS OF NUMERICAL ANALYSIS

All the presented results were obtained for currents

$$I_S = 1.5 I_{\text{lim}}.$$

Structure of Diffusion Layer

From the analysis of the charge density distribution (Fig. 2a) it follows that the diffusion layer consists, in the most general form, of the electroneutrality region (ENR) and the space charge region (SCR).

The data in Fig. 2 show:

(1) The electroneutrality region begins from $x = 0$ and continues to a certain point x_c (in Fig. 2a, $x_c \approx 0.7$). Since the relations $j_3 \approx \text{const}$ and $j_4 \approx \text{const}$ hold everywhere in the ENR with the exception of the recombination region (Fig. 5), it follows from Eqs. (16) and (18) that the equilibrium condition of the dissociation/recombination reaction $k_W - C_3 C_4 = 0$ is fulfilled everywhere in the ENR except the RR (Fig. 3).

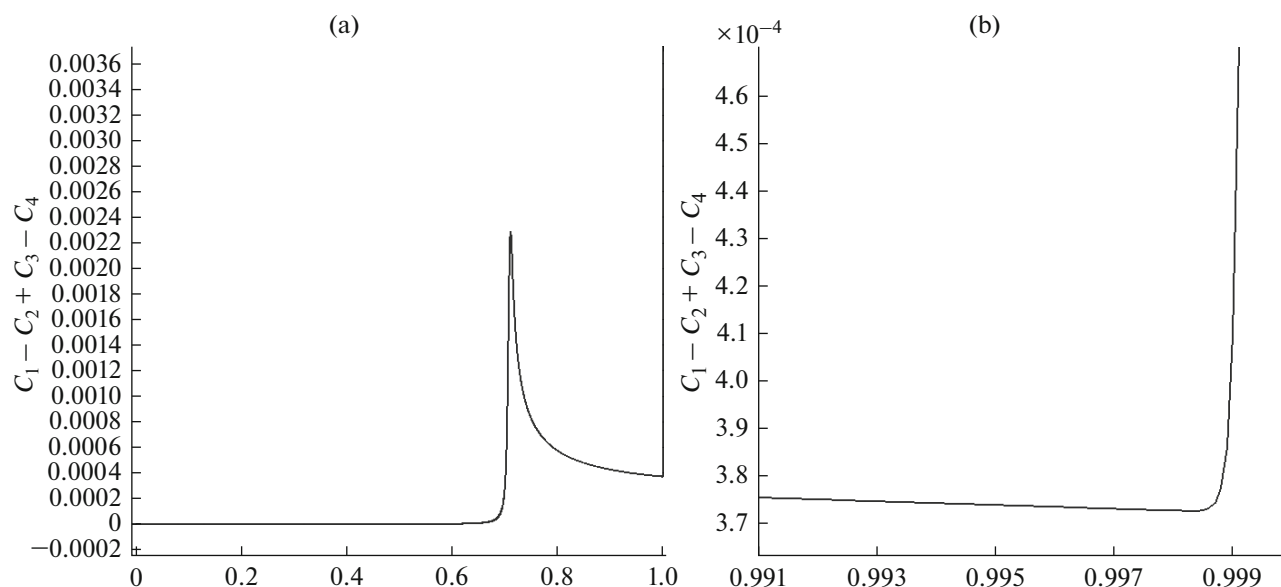


Fig. 2. Distribution of charge density at $C_{10}^{(u)} = 1 \text{ mol/m}^3$, $C_{30} = 0.01 \text{ (0.01 mol/m}^3\text{)}^1$ (as a consequence, pH 5), $C_{3m} = 10^{-6} \text{ (10}^{-6} \text{ mol/m}^3\text{)}$, and $j_{4m} = -0.4 \text{ (-2} \times 10^{-5} \text{ mol m}^{-2} \text{ s}^{-1}\text{)}$: (a) general view and (b) enlarged at $x = 1$.

(2) The space charge region starts approximately from x_c and continues to the end of the segment $x = 1$. The SCR width $(1 - x_c)$ in Fig. 2 is approximately 0.3; the SCR (Figs. 2b, 5, 6) consists of the extended (main) part $x \in (x_c, x_q)$, where all the sought functions change relatively slowly, and the region $x \in (x_q, 1]$ (in Fig. 2, $x_q \approx 0.9995$) located near the cation-exchange membrane surface, where the functions vary exponentially.

The extended SCR arises only in the case of over-limiting current densities, its dimensions are finite at $\varepsilon \rightarrow 0$ and are determined by how much the current density exceeds the limiting diffusion current density [25]. The thickness of this region in a binary electrolyte can be calculated according to the approximate formula [16, (6.124)]:

$$\delta_{\text{SCR}} = \frac{\varepsilon I_s}{2C_{1s}^2},$$

where C_{1s} is the minimum value of cation concentration. It is in this region that the noncatalytic reaction of the splitting of water molecules occurs.

The region $x \in (x_q, 1]$ is half the quasi-equilibrium electrical double layer (EDL). The existence and dimensions of the region $(x_q, 1]$ are determined by the presence of a small parameter ε and by the boundary conditions at $x = 1$. The thickness of this region is

defined by the formula [20]: $1 - x_q \approx \sqrt{\frac{2\varepsilon}{C_{1s}}} \sim \sqrt{\varepsilon}$. Calculations show that the influence of the boundary conditions $C_1|_{x=1} = C_{1m}$ and $C_3|_{x=1} = C_{3m}$ is appreciable only in the region of the quasi-equilibrium EDL. Therefore, in subsequent calculations, these boundary values are unchanged (in dimensionless form) and are taken to be $C_{1m} = 1$ and $C_{3m} = 10^{-6}$. In contrast to these boundary conditions, the condition

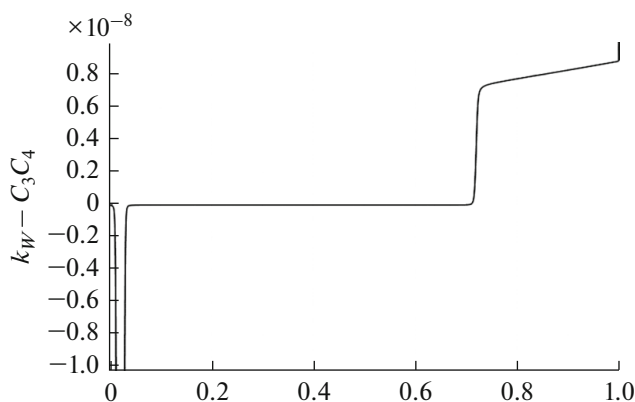


Fig. 3. Plot of the function $k_W - C_3C_4$ characterizing the deviation from equilibrium at $C_{10}^{(u)} = 1 \text{ mol/m}^3$, $C_{30} = 0.01 \text{ (0.01 mol/m}^3\text{)}$ (pH 5), $C_{3m} = 10^{-6} \text{ (10}^{-6} \text{ mol/m}^3\text{)}$, and $j_{4m} = -0.4 \text{ (-2} \times 10^{-5} \text{ mol m}^{-2} \text{ s}^{-1}\text{)}$.

¹ Hereinafter, the values of the quantities are given in both dimensionless and dimensional form (in parentheses) for convenience.

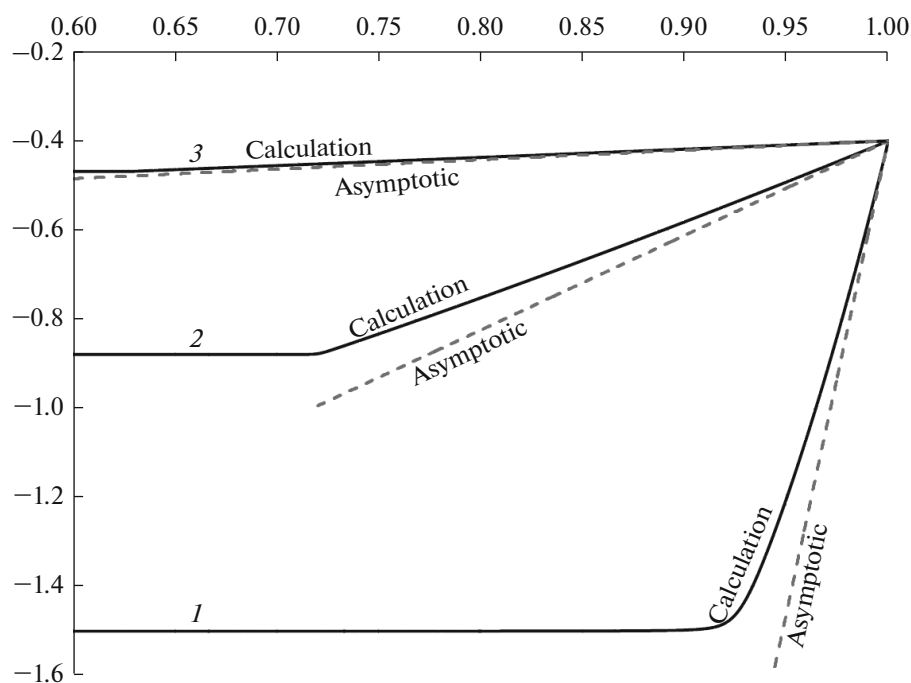


Fig. 4. Comparison of the numerical and asymptotic solutions for OH^- fluxes at various values of $C_{10}^{(u)}$: (1) $C_{10}^{(u)} = 0.1 \text{ mol/m}^3$ (pH 6), (2) $C_{10}^{(u)} = 1 \text{ mol/m}^3$ (pH 5), and (3) $C_{10}^{(u)} = 10 \text{ mol/m}^3$ (pH 4).

$\left(\frac{dC_4}{dx} + C_4 E\right)_{x=1} = -j_{4m}$ affects the solution over the entire interval $(0, 1)$; that is, throughout the entire diffusion layer. In the region of quasi-equilibrium EDL, water molecules dissociate, but the size of this region is so small (in Fig. 3, the length of this zone is approximately 5×10^{-4} ($5 \times 10^{-8} \text{ m}$)). Because of this, as Simons rightly noted [12], the dissociation of water molecules in the region of quasi-equilibrium EDL does not provide the fluxes of hydrogen and hydroxyl ions observed in practice.

(3) Between ENR and SCR, near the point $x = x_c$, there is an intermediate region (IR) having a small width (Fig. 2a), in which the solutions pass from the electroneutrality region $[0, x_c)$ and the space-charge region $(x_c, 1]$. In the intermediate region, the charge density has a local maximum, the equilibrium condition is no longer satisfied (Fig. 3), and the derivatives of hydrogen and hydroxyl fluxes change very rapidly.

Analysis of the Reactions of Dissociation of Water Molecules and Recombination of Hydrogen H^+ and Hydroxyl OH^- Ions

The noncatalytic reaction of water dissociation in the extended space charge region. The results of numerical and asymptotic solutions show that everywhere with the exception of the reaction zone and the SCR, one of the fluxes is almost zero and the other is constant.

In the SCR, both fluxes are nonzero, and they increase almost linearly in the extended SCR, reaching significant values. For the analysis, it is sufficient to consider the fluxes of OH^- ions (Figs. 4, 5). The inclined segments in Fig. 4 indicate that the dissociation of water molecules occurs intensively in the main part of the SCR, with its intensity being due to high values of electric field strength in the extended SCR (Fig. 6a). Indeed, because of the large values of the electric field strength in the main part of the SCR, the concentration of hydrogen and hydroxyl ions becomes so small (Fig. 7) that the equilibrium condition in the extended SCR is violated; moreover, $k_w \gg C_3 C_4$.

For example, for $C_{10}^{(u)} = 1 \text{ mol/m}^3$, $C_{30} = 0.01$ (0.01 mol/m^3), pH 5, $I_S = 1.5I_{\text{lim}}$, $j_{4m} = -0.4$ ($-2 \times 10^{-5} \text{ mol m}^{-2} \text{ s}^{-1}$), and $C_{3m} = 10^{-6}$ (10^{-6} mol/m^3), we have $k_w = 10^{-8}$, and $C_3 C_4 < 10^{-9}$; therefore

$\varepsilon D_3 \frac{d}{dx} j_3 \approx ak_w$, i.e., $j_3 \approx \frac{ak_w}{\varepsilon D_3} (x - x_c)$, where x_c is the left boundary of the SCR. Thus, the flux j_3 is sufficiently large already at $\frac{ak_w}{\varepsilon D_3} (x_q - x_c) \sim 1$. From the

equation $I_w = D_3 j_3 - D_4 j_4 = \text{const}$, it follows that the flux j_4 is also sufficiently large, with

$j_4 \approx j_{4m} + \frac{ak_w}{\varepsilon D_4} (x - 1)$, as reflected in Fig. 4.

From Fig. 4 it follows that the values of the fluxes are quite large at low salt concentrations ($C_0^{(r)} \ll 1 \text{ mol/m}^3$), at the same time, at high salt concentrations ($C_0^{(r)} \gg 1 \text{ mol/m}^3$), we get $\frac{d}{dx} j_3 \approx \frac{d}{dx} j_4 \approx 0$. Hence, it follows that at high salt concentrations, the equilibrium condition $k_w = C_3 C_4$ is satisfied in the extended SCR and the role of noncatalytic water dissociation in the extended SCR becomes unnoticeable.

Thus, the lower the salt concentration in the bulk of the solution, the more intense is the noncatalytic dissociation of water in the extended SCR.

The recombination reaction of hydrogen and hydroxyl ions. In the general case, the reaction layer in which hydrogen H^+ and hydroxyl OH^- ions recombine (recombination region, RR) is located in the region of electroneutrality. In the recombination region, the fluxes change exponentially (Fig. 5). Note that the electric field strength undergoes a sharp change in the RR, as shown in Fig. 6. However, the concentrations do not change drastically in the recombination region.

The recombination region shown in Fig. 5 lies in the interval (0.015, 0.03) with the “center” near the point $x_r \approx 0.02$. The width of the RR is small in comparison with the width of the diffusion layer; in Fig. 5a, it is approximately 1.3×10^{-3} ($1.3 \times 10^{-7} \text{ m}$), which is 1.3% of the diffusion layer width. In this connection, the reaction layer in many cases can be represented by the point (x_r), the RR “center.” The recombination point is defined by the equality $C_3(x_r) = C_4(x_r)$. Since x_r lies in the ENR, it follows from $C_3(x_r) = C_4(x_r)$ that $C_1(x_r) = C_2(x_r)$.

The ratio of catalytic and noncatalytic routes of dissociation of water molecules. In order to consider the ratio between the catalytic and noncatalytic routes of dissociation of water molecules, it is necessary to distinguish the noncatalytic dissociation, assuming that $j_{4m} = 0$ and compare the results with $j_{4m} \neq 0$, for example, with $j_{4m} = -0.2$ ($-10^{-5} \text{ mol m}^{-2} \text{ s}^{-1}$).

First of all, it should be noted that the fluxes differ substantially in the quasi-equilibrium SCR, as shown in Fig. 5; this difference is naturally associated with the difference in the boundary condition at the interface.

In the other regions, the difference in magnitude between the fluxes is up to 10–20%. Thus, it can be concluded that the main contribution to the dissociation of water molecules at low salt concentrations is made by noncatalytic dissociation. However, as the salt concentration increases, the contribution of noncatalytic dissociation decreases and catalytic dissociation begins to dominate.

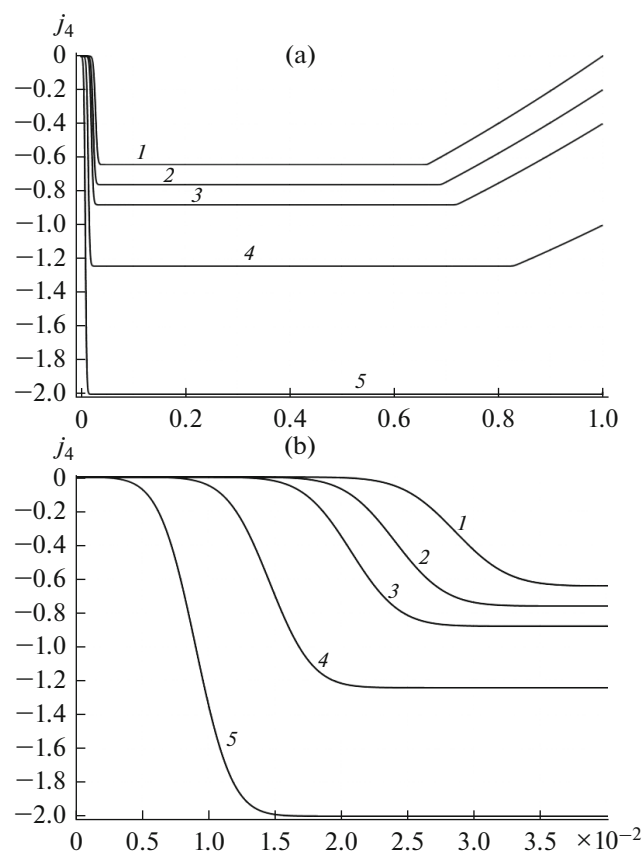


Fig. 5. Distribution of fluxes (a) over the entire thickness of the diffusion layer and (b) near $x = 0$ at $C_{10}^{(u)} = 1 \text{ mol/m}^3$ and $C_{30} = 0.01$ (0.01 mol/m^3) (pH 5): (1) $j_{4m} = 0$, (2) $j_{4m} = -0.2$ ($-10^{-5} \text{ mol m}^{-2} \text{ s}^{-1}$), (3) $j_{4m} = -0.4$ ($-2 \times 10^{-5} \text{ mol m}^{-2} \text{ s}^{-1}$), (4) $j_{4m} = -1.0$ ($-5 \times 10^{-5} \text{ mol m}^{-2} \text{ s}^{-1}$), and (5) $j_{4m} = -2.0$ ($-10^{-4} \text{ mol m}^{-2} \text{ s}^{-1}$).

Analysis of Electric-Field Strength and Concentration Distributions

Sokirko and Kharkats [21] assumed that at a point x_r in the recombination region, the electric field strength can exhibit a “spike.” However, Fig. 6b shows that there is no “spike” at all. In the RR, there is a transition from one solution to another, which is due to the fact that the electric field strength to the left of the point x_r is determined by the flux of hydrogen ions and the field to the right is determined by the flux of hydroxyl ions, and they are not equal in absolute value because of the difference in their diffusion coefficients.

As can be seen from Fig. 7a, the presence of the OH^- ion leads to the fact that it begins to displace and replace the Cl^- anion already in the electroneutrality region, and the Cl^- concentration in the space charge region is zero with great accuracy because of the assumption on ideal selectivity of the cation-exchange

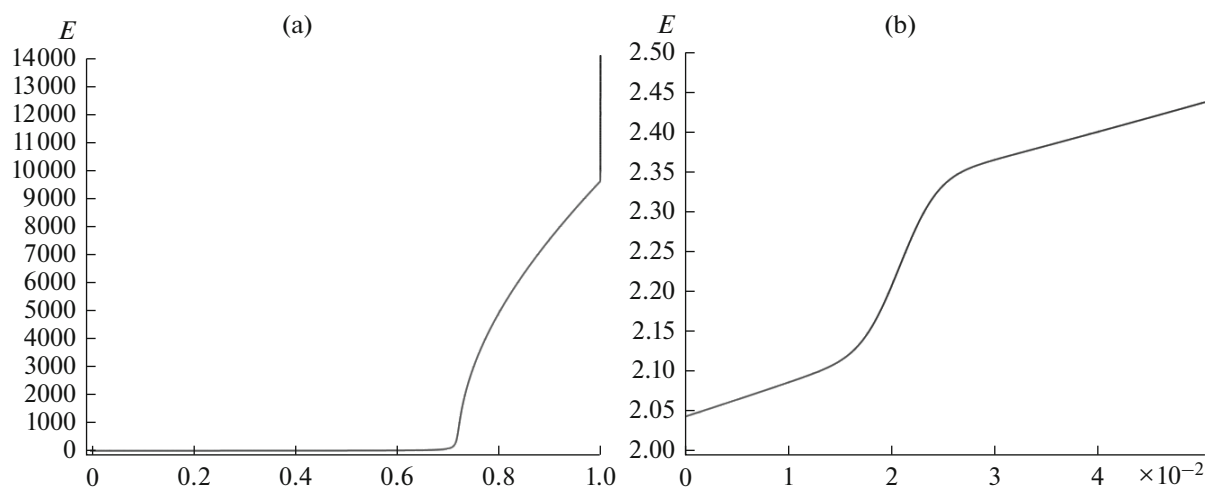


Fig. 6. Distribution of electric field strength at $C_{10}^{(u)} = 1 \text{ mol/m}^3$ and $C_{30} = 0.01$ (0.01 mol/m³) (pH 5): (a) general view and (b) enlarged in the recombination area.

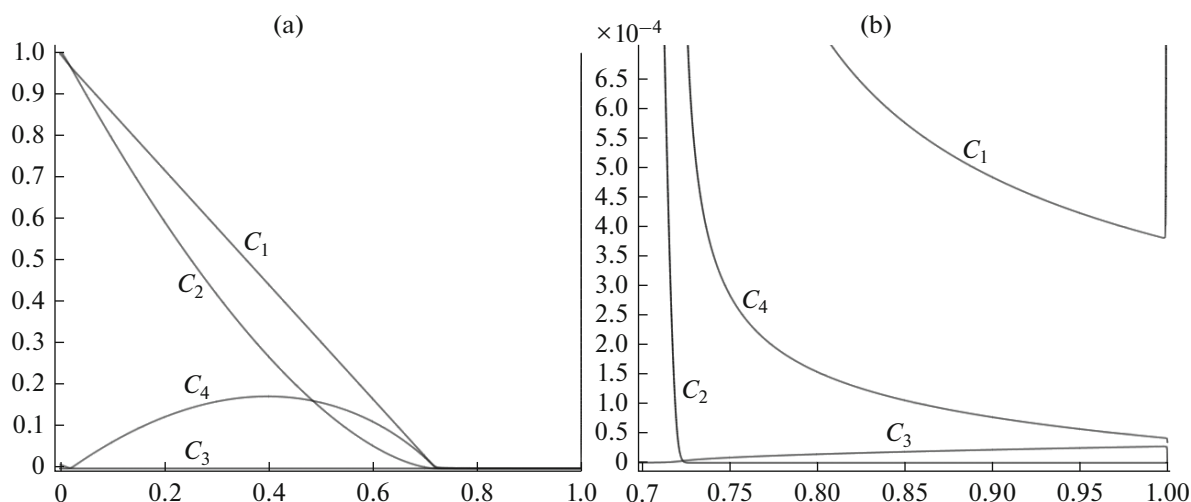


Fig. 7. Distribution of concentrations at $C_{10}^{(u)} = 1 \text{ mol/m}^3$ and $C_{30} = 0.01$ (0.01 mol/m³) (pH 5): (a) general view and (b) enlarged near $x = 1$.

membrane. Concentrations in the recombination region do not vary dramatically. The results in the electroneutrality region are consistent with the published data [21] obtained at underlimiting current densities.

CONCLUSIONS

The structure of the diffusion layer under overlimiting current regimes has been established. It has been shown that intense noncatalytic splitting of water molecules occurs in the extended space charge region, which has finite dimensions, at small salt concentrations.

The fluxes of hydrogen and hydroxyl ions have significant values in the extended SCR and are distrib-

uted according to an almost linear law. This is due to the fact that the electric field strength in the SCR reaches so large values that the product of concentrations of hydrogen and hydroxyl ions becomes significantly less than the equilibrium constant; i.e., the equilibrium condition $C_3C_4 = k_w$ of the dissociation/recombination reaction is violated in favor of dissociation.

The size of the recombination region has been determined, which is much smaller than the width of the diffusion layer (it is on the order of 1 to 2%), and therefore, the RR can be replaced by the "recombination point" in many cases.

It has been shown that the equilibrium condition is satisfied everywhere in the diffusion layer, except the recombination and space charge regions. Wherever

the equilibrium condition is satisfied, the concentration of either hydrogen or hydroxyl ions can be considered with good accuracy to be zero; i.e., it can be assumed that these ions are not present simultaneously.

The revealed trends in the influence of the dissociation/recombination reaction on the transport of salt ions make it possible to explore the possibility of the emergence or enhancement of gravitational convection due to the dissociation/recombination reaction of water molecules, which opens the opportunity of intensifying the transport process of salt ions.

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